DR. SIDDHARTH MEHTA

Principal AI & NLP Research Scientist | Deep Learning Architect

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EXECUTIVE SUMMARY

Pioneering Deep Learning Researcher with 7+ years of expertise leading multimodal foundation models, document intelligence systems, and transformer-based LLM architectures. Author of 12 peer-reviewed papers in NeurIPS, CVPR, and ACL. Expert in PyTorch, Distributed Training, and End-to-End Multimodal Vision-Language Systems.

TECHNICAL SKILLS & COMPETENCIES

· Deep Learning: PyTorch, TensorFlow, Torchvision, CNN, Bi-LSTM, Autoencoders, VAE, GANs

· NLP & Transformers: BERT, RoBERTa, GPT, Word2Vec, GloVe, TF-IDF, POS Tagging, NER, N-Grams

· Computer Vision: YOLOv8, ResNet50, VGG16, LayoutLMv3, OpenCV, PyMuPDF, OCR Engines

· Cloud & APIs: FastAPI, Docker, Kubernetes, MongoDB, Redis, Streamlit, Groq LLM Cloud API

PROFESSIONAL EXPERIENCE

Principal AI Scientist | Cortex Intelligence Labs (2023 - Present)

· Spearheaded the development of multimodal document classification systems processing 2M+ docs/month.

· Reduced OCR error propagation by 38% using custom Convolutional Autoencoder (CAE) denoising filters.

Senior Machine Learning Engineer | DeepVision AI (2020 - 2023)

· Designed Bi-LSTM sequence labeling pipelines achieving 96.4% F1-score for tabular invoice parsing.

· Implemented multimodal late-fusion architectures combining visual ResNet embeddings and BERT text features.

DR. SIDDHARTH MEHTA · EDUCATION & PUBLICATIONS

EDUCATION & ACADEMIC CREDENTIALS

Ph.D. in Computer Science & Artificial Intelligence (2016 - 2020)

Indian Institute of Science (IISc), Bangalore | Focus: Multimodal Representation Learning

B.Tech in Computer Engineering (2012 - 2016)

National Institute of Technology (NIT) | First Class with Distinction (GPA: 9.4/10)

KEY RESEARCH PUBLICATIONS

1. Mehta, S., et al. (2024). 'Confidence-Weighted Late Fusion in Document Understanding.' NeurIPS.
2. Mehta, S., et al. (2022). 'Variational Autoencoders for Scanned Document Inpainting.' CVPR.
3. Mehta, S., et al. (2021). 'Bi-LSTM Entity Extraction over Dense Tabular Layouts.' EMNLP.